

Structural Self-Energy Expansion (SSEE)

A minimal-parameter dark energy framework from φ and π

Mike Edison Almeida Vallejo [ORCID: 0009-0008-2195-7836](https://orcid.org/0009-0008-2195-7836)
mike.almeida1721@gmail.com github.com/mikealmeida1721/SSEE

Seeking arXiv endorsement in astro-ph.CO (or gr-qc)

Core idea in one sentence

The core cosmological background — w_0 , w_a , H_0 , Ω_m , n_s — is derived *algebraically* from the golden ratio $\varphi = (1 + \sqrt{5})/2$ and π with **zero fitted dimensionless parameters at the background level**. The kineticity α_K^{field} of the dark-energy sector is a working EFT proposal, subject to full Boltzmann validation. The dark-sector ansätze that previously appeared here (m_φ , ξ) were retracted in 2026-08 together with the second matter sector, and their register entries (OP-9 through OP-12) close by dissolution; the MIRA matter mechanism (OP-8) had already dissolved under the ω_m -direct reframe. The remaining open entries are listed in `OPEN_PROBLEMS.md`.

Key algebraic predictions

Quantity	SSEE algebraic value	Observed / constraint	Agreement
w_0	$-T_r/M_v = -0.8400$	DESI DR2: -0.838 ± 0.055	0.24σ
w_a	$-P_{sc}/I_g = -0.669975$	DESI DR2: -0.631 ± 0.226	0.17σ
H_0^{alg}	$3(\varphi + \pi)^2 = 67.96214 \text{ km/s/Mpc}$	Planck 2018: 67.36 ± 0.54	1.11σ
$\Omega_{m,\text{CMB}}$	$\omega_m/h^2 = 0.308881$ (ω_m -direct)	Planck 2018: 0.3153 ± 0.0073	0.88σ
r_d	CAMB (ω_m -direct): 147.17 Mpc	Planck 2018: $147.09 \pm 0.26 \text{ Mpc}$	0.32σ
H_0^{glob}	$H_0^{\text{SHOES}}(1 - f_{\text{scr}}) = 68.13 \text{ km/s/Mpc}$	$3(\varphi + \pi)^2 = 67.96214$	0.17σ
n_s	$1 - \varphi^{-7} = 0.965558$	Planck PR4: 0.9649 ± 0.0042	0.16σ
r	$\varphi^{-10} = 0.008131$	BK18: $r < 0.036$	Prediction
α_K^{field}	$3\Omega_{\text{DE}}(1 + w_\varphi) = 0.072567$	Euclid forecast sensitivity	< 0.1
S_8	0.8273 (predicted; <i>no</i> free cosmological parameter)	KiDS-Legacy: 0.8265 ± 0.0176	0.05σ

Ten-paper series (all reproducible)

I. Derived Algebraic Background

- **Paper 1 (Framework)**: Algebraic derivation of geometric constants; α -attractor embedding ($\alpha = \varphi^4/3$). [35 pp]
- **Paper 4 (Nine Sovereignties)**: CMB observables ($\ell_1, \ell_2, \ell_3, \theta_*, A_s, n_s, \tau, r$) bounded algebraically. [19 pp]

II. Tested Against Data

- **Paper 2 (MCMC)**: 100 walkers $\times$ 25 000 steps vs. DESI DR2 + Planck + 7 clusters; $\Delta\text{BIC} = -6.43$ (SSEE favoured, $k_{\text{SSEE}} = 2$ vs. $k_{\Lambda\text{CDM}} = 3$), total matter density in the background geometry. [31 pp]
- **Paper 3 (CMB Confrontation)**: CAMB ω_m -direct ($\Omega_m = 0.308881$); Cobaya full `plik` MCMC; $\Delta\text{BIC} = -32.9$ (canonical; -26.21 in the `plik_lite` point estimate with $\{A_s, \tau\}$ fitted and $N = 669$ measured from the likelihood), SSEE favoured under $k = 2$ vs $k = 6$ model comparison. [26 pp]
- **Paper 6 (Growth sector, against raw $\xi_\pm$)**: retracts an earlier two-sector φ -DM proposal and its particle — the subtraction defining the second sector mixed a measured density with an equation-of-state number, and the S_8 tension motivating it exists only in the compressed

statistic, whose reduction assumes Λ CDM. Against the **KiDS-Legacy** release (357 raw points; Wright et al. 2025, arXiv:2503.19441 — *published sixteen months before this analysis: no temporal priority is claimed*) the shear asks for $\log(10^{10} A_s) = 3.0255 \pm 0.0396$ against the 3.04483 that the CMB fixes under the *same* algebraic background: 0.49σ (Planck background, held equally rigid: 1.01σ). Fixing A_s to that CMB value therefore costs nothing — $\chi^2 = 417.97$ with **eight free parameters, all nuisance, none cosmological**, against 418.34 with A_s free ($\Delta\text{BIC} = +6.24$) and 418.95 for the Λ CDM-background control (+6.86). $S_8 = 0.8273$ is then predicted, not fitted, against 0.8265 ± 0.0176 measured: 0.05σ . The three χ^2 span 1.0, so the shear does *not* discriminate between the models; what separates them is the parameter count. (Earlier, against KiDS-1000: $S_8 = 0.7555 \pm 0.0192$ with A_s free, matching the Λ CDM control 0.7571 ± 0.0194 run through the same pipeline with four more free parameters.) [17 pp]

III. Phenomenological Extensions & Open Directions

- **Paper 5 (Israel–Stewart Perturbations)**: Causal bulk viscosity; $c_s^2 = 0$ as the minimal-viscosity solution (OP-21); $\gamma_{\text{IS}} = 0.5504 \approx \gamma_{\Lambda\text{CDM}}$. [31 pp]
- **Paper 7 (Canonical EFT Proposal)**: the action is a minimally coupled phantom condensate, $K(X) = c_1 X + c_2 X^2$, with $\alpha_T = \alpha_M = \alpha_B = 0$ and bare-field $\alpha_K^{\text{field}} = 3 \Omega_{\text{DE}}(1 + w_\varphi) = 0.072567$ algebraic. *Two structures that earlier versions carried were withdrawn in 2026-09*, with the reasons recorded in the paper because they are themselves results: the exponential potential (its exponent was a tautology of the tracker relation) and the conformal dark coupling $\beta_c = -\text{AURA}$, whose apparent 0.2% agreement was an artefact of a normalisation error in the saturation — corrected, the value is -2.194210 , some 45% away. Neither structure appears in the action. [17 pp]

IV. Strong-Field Regime, Hubble Tension & UV Completion

- **Paper 8 (Strong Gravity)** [*draft; its lensing prediction is conditional on EFT limit (b) since the φ -DM sector was retracted*]: Disformal null geodesic analysed in two limits: alternative MOND-like (deflection $\sqrt{\text{AURA}} \approx 1.9995$, numerically close to $\mathcal{M} = \text{AURA}/2$ at 0.03% but algebraically distinct) and the canonical limit with EFT $\alpha_B = \alpha_M = \alpha_T = 0$ (residual fifth-force $\sim 10^{-15}$ at kpc, observable lensing recovers GR); selective coupling satisfies solar-system tests. Current SLACS/BELLS data favour the canonical limit (OPEN_PROBLEMS.md OP-13). A dimensional defect in the Vainshtein-radius equation of this paper is open and under repair (OP-4, reopened 2026-09); it does not enter the canonical-limit prediction, which rests on ω_c and on $\alpha_B = \alpha_M = 0$. [19 pp]
- **Paper 9 (Hubble Tension)**: Algebraic local screening fraction $f_{\text{scr}} = (\pi - \varphi)/(\varphi + \pi)^2 \approx 0.067$ applied to the measured SH0ES rate 73.04 ± 1.04 returns a global background $H_0^{\text{glob}} = 68.13$ km/s/Mpc, 0.17σ from the pure algebraic number $3(\varphi + \pi)^2 = 67.96214$ — an algebraic prediction with no fitted parameters (2026 reframe; the earlier CMB-mapped $H_0^{\text{MIRA}} = 67.04$ is superseded). [20 pp]
- **Paper 10 (UV Completion)**: Higher-derivative operator $K(X) = X/\text{KAL}_0 + X^2/M^4$ with algebraic cutoff $M = 9.68$ meV; connects the dark-energy EFT to the inflationary α -attractor. [16 pp]

Falsifiability (explicit)

- $S_8 = 0.8273$ — with the background fixed by the algebra *and* A_s fixed to the value the CMB gives under that same background, the growth sector has no free cosmological parameter left: cosmic shear placing S_8 more than 3σ away falsifies it outright, with nowhere to absorb the discrepancy (KiDS, DES, Euclid).
- $\log(10^{10} A_s)$ — the amplitude the shear asks for must keep agreeing with the one the CMB fixes. KiDS-1000 asked for 2.863 ± 0.051 (3.5σ away) and KiDS-Legacy for 3.0255 ± 0.0396 (0.49σ); the

two releases differ from each other by 2.5σ . A future release returning to the lower value reopens the sector.

- $\alpha_K^{\text{field}} \approx 0.073$ — Euclid weak lensing will probe this kinematicity limit.
- $r = \varphi^{-10} = 0.008131$ — LiteBIRD (~ 2032) will confirm or refute the tensor-to-scalar ratio.
- A $> 3\sigma$ deviation of w_0 from -0.840 would falsify the corresponding background sector of the framework.

What I am asking for

An arXiv endorsement in `astro-ph.CO` (or `gr-qc`) to submit Papers 1–3 as the first tranche. All code, data, and L^AT_EX sources are public at github.com/mikealmeida1721/SSEE under Apache-2.0.

I am happy to answer any technical questions or share the PDFs directly. Thank you for considering this request.

Code: Python (emcee, CAMB, CLASS), all reproducible in <30 min. **Data:** DESI DR2 (public), Planck PR4 (public), KiDS-1000 and KiDS-Legacy (public).